Predicting College Application Volume Using Characteristics of U.S. Colleges

STAT-387 Intro to Statistical Learning

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Overview

Goals of the Project

This report will be analyzing different models using the College dataset from the R package "ISLR2" with the goal of giving the best prediction for the number of applications received using all the other variables in the data set. The College dataset contains statistics on 777 US colleges from the 1995 US News and World Report, which includes institutional characteristics that range from enrollment figures and tuition costs to faculty qualifications and graduation rates. Understanding what drives application volume is a meaningful problem, as application counts serve as a proxy for a college's visibility, selectivity, and overall demand among prospective students.

As well as this, there will be an exploration of the relationships between the other factors that contribute to the overall number of applications received by a college and how these various factors intermingle. Some predictors, such as acceptance rate and enrollment figures, are closely tied to institutional size and are expected to dominate prediction, while others, such as faculty qualifications and out of state tuition, may reveal subtler patterns in applicant behavior. This analysis will ultimately focus on which of the models best predicts the number of applications received. We will then recommend which models to use for analyzing this data, with model performance assessed mainly by mean squared error.

Project Content

The analysis begins with an exploratory data analysis consisting of graphical plots, numerical summary tables, and a correlation matrix to help with understanding the dataset and the relationship between variables. Exploratory Data Analysis (EDA) reveals important distributional characteristics of the response variable. These include a notable right-skewed distribution, mainly due to a small number of institutions with high volumes. In addition, strong correlations are seen among several predictors related to size, which will influence later modeling decisions.

Next, different models are fit to the data, specifically a linear model using least squares as well as a regression tree. Two versions of each model were created in order to compare the effect of certain predictors, one using the full set of available variables and a second reduced model that excludes the most dominant size-related predictors, which will be explained further in later sections. This comparison allows for a clearer view of how the remaining predictors behave when the strongest signals are removed from the analysis. Then a bagging approach and the random forest method were used to train multiple regression models with increasing B, and the MSE of every model was compared to determine the best method. Both ensemble methods are run at B = 500 and B = 1000 trees to examine whether additional computational investment translates into meaningful gains in predictive accuracy, and variable importance plots are used throughout to interpret which predictors contribute most to each model.

Exploratory Data Analysis

Overview of Variables

The College dataset from the ISLR2 library contains statistics for US colleges from US News and World Report from 1995. The dataset includes 18 variables; Apps, the number of applications received, serves as the response variable, while the remaining 17 variables serve as predictors. Table 1 below provides a description of each variable.

Variable	Description
Apps	Number of applications received (response variable)
Accept	Number of applications accepted
Private	Factor: No = public university, Yes = private university
Enroll	Number of new students enrolled
Top10perc	Percentage of new students from top 10% of high school class
Top25perc	Percentage of new students from top 25% of high school class
F.Undergrad	Number of full-time undergraduate students enrolled
P.Undergrad	Number of part-time undergraduate students enrolled
Outstate	Out-of-state tuition
Room.Board	Room and board costs
Books	Estimated book costs
Personal	Estimated personal spending
PhD	Percentage of faculty members with a PhDs
Terminal	Percentage of faculty members with a terminal degree
S.F.Ratio	Student-to-faculty ratio
Perc.alumni	Percentage of alumni who donate to the college
Expend	Instructional expenditure per student
Grad.Rate	Graduation Rate

Table 1. *Variable Descriptions for the College Dataset*

Table 2 provides a summary of the statistical characteristics of the eighteen variables within the College dataset. The continuous predictors are characterized by their mean, median, standard deviation, minimum, and maximum values. A significant finding is the pronounced right skew observed in the variables Apps, Accept, and F.Undergrad, as indicated by means that considerably exceed their corresponding medians. This skewness is a crucial factor in the model fitting process, and its implications are explored in greater detail in the subsequent section, which focuses on the Visualization of Relationships.

Variable	Mean	Median	Std. Dev	Min	Max
Apps	3002	1558	3870	81	48094
Accept	2019	1110	2452	72	26330
Private	Yes = 565 No = 212	—	—	—	—
Enroll	780	434	930	35	6392
Top10perc	27.6	23.0	17.6	1	96
Top25perc	55.8	54.0	19.8	9	100
F.Undergrad	3700	1707	4855	139	31643
P.Undergrad	855	353	1552	1	21836
Outstate	10441	9990	4024	2340	21700
Room.Board	4358	4200	1096	1780	8124
Books	549	500	165	96	2340
Personal	1341	1200	677	250	6800
PhD	72.7	75.0	16.3	8	103
Terminal	79.7	82.0	14.7	24	100
S.F.Ratio	14.1	13.6	3.96	2.5	39.8
Perc.alumni	22.7	21.0	12.4	0	64
Expend	9660	8377	5221	3186	56233
Grad.Rate	65.5	65.0	17.2	10	118

Table 2. *Summary Statistics for the College Dataset. Statistics are based on n = 777 colleges. For the Private variable, counts are reported in place of numerical summaries. Mean and standard deviation for Apps and Accept are notably large relative to their medians, indicating right skew driven by a small number of high-volume institutions.*

Visualization of Relationships

Before creating models with the dataset, the variables of the dataset were compared using a variety of plots in order to notice patterns in the relationship that were worth exploring. Since Apps was the variable of interest, a histogram was created showing the distribution of the number of applications across all colleges in the dataset.

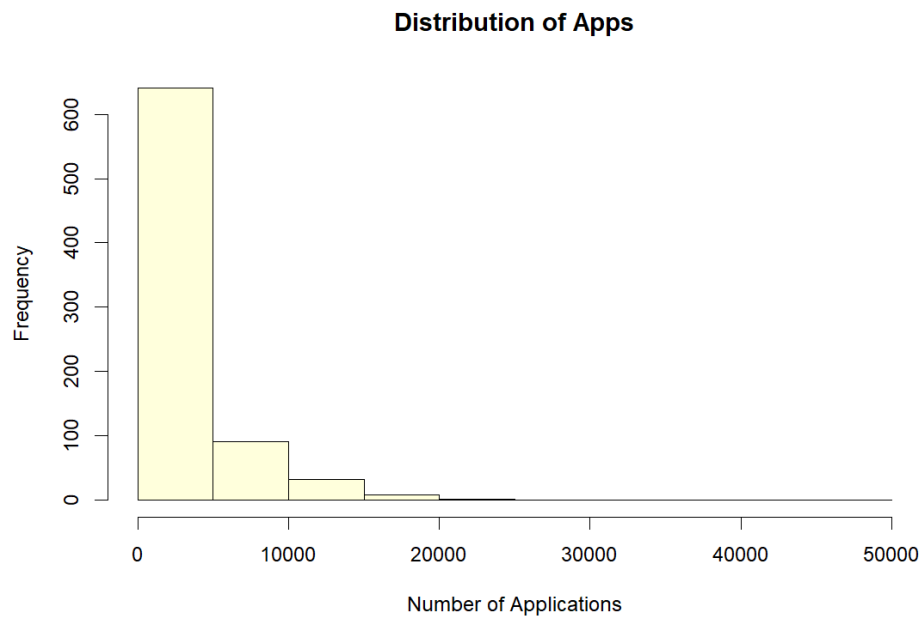

Figure 1. *Distribution of Apps. Number of Applications across colleges.*

Looking at this plot, it is clear that there is a small number of colleges that have a large number of applications that skew the entire dataset, while over 600 colleges have lower than 5000 applications. The data being skewed is significant, as it will come up while fitting models and influence the steps moving forward. Next, scatterplots were created to determine the correlations between Apps and the two biggest variables related to College applications: Number of Acceptances and Number of Enrollments. These are different from Applications as applicants get accepted, then choose to enroll.

To view the relationship between all variables at once, a correlation matrix was created. This is a visual plot showing the strength and direction of multiple variables, using the PerformanceAnalytics library in R. By looking at the correlation matrix, it becomes easier to identify strongly related predictors and determine which variables may have meaningful linear relationships that can lead to further analysis

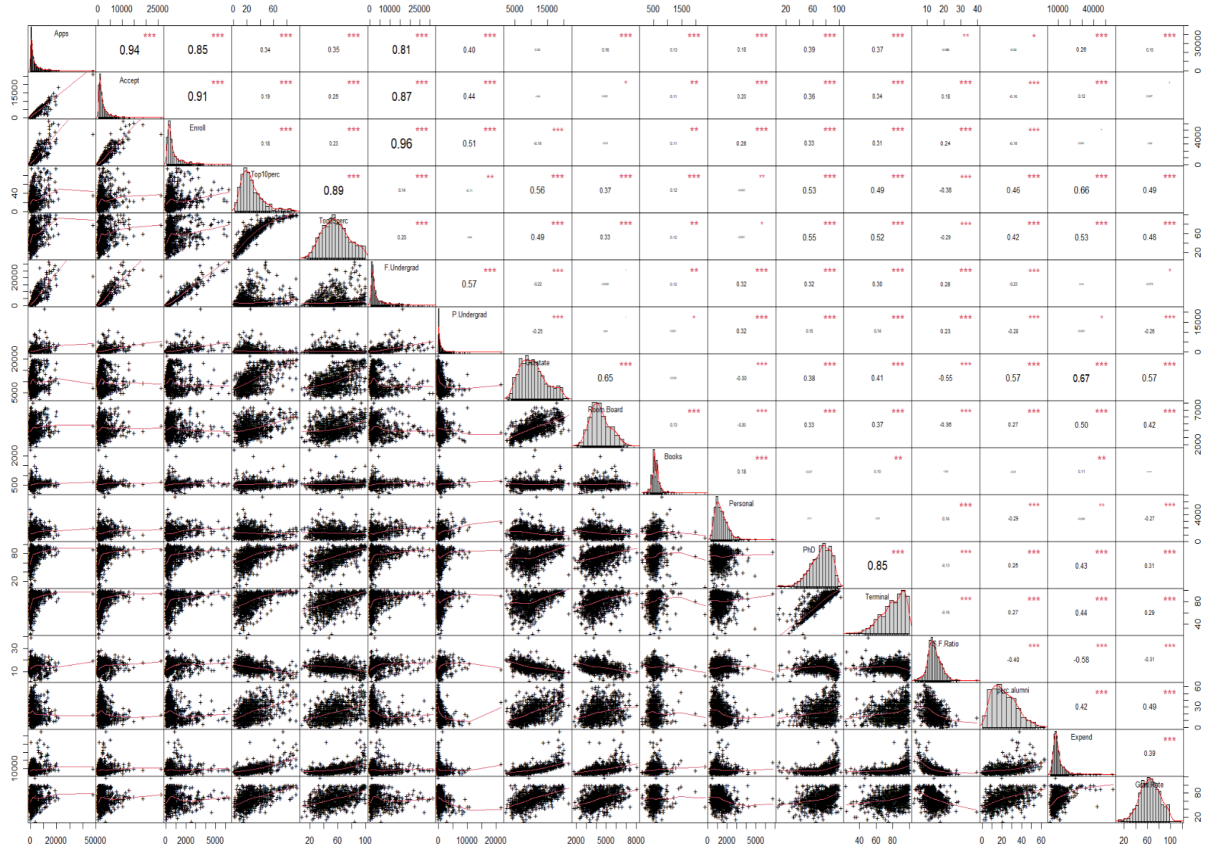

Figure 2. *Correlation Matrix of College dataset. Variables to note are Accept, Enroll, and F.Undergrad. These are to be expected as they are closely related to the population of a college and are all highly correlated. Other clusters to note are Top10perc and Top25 perc being the top 10 and top 25 percent of students from their high schools, as well as PHD and Terminal which are faculty with PHDs and terminal degrees, the highest qualification in their specific field of study.*

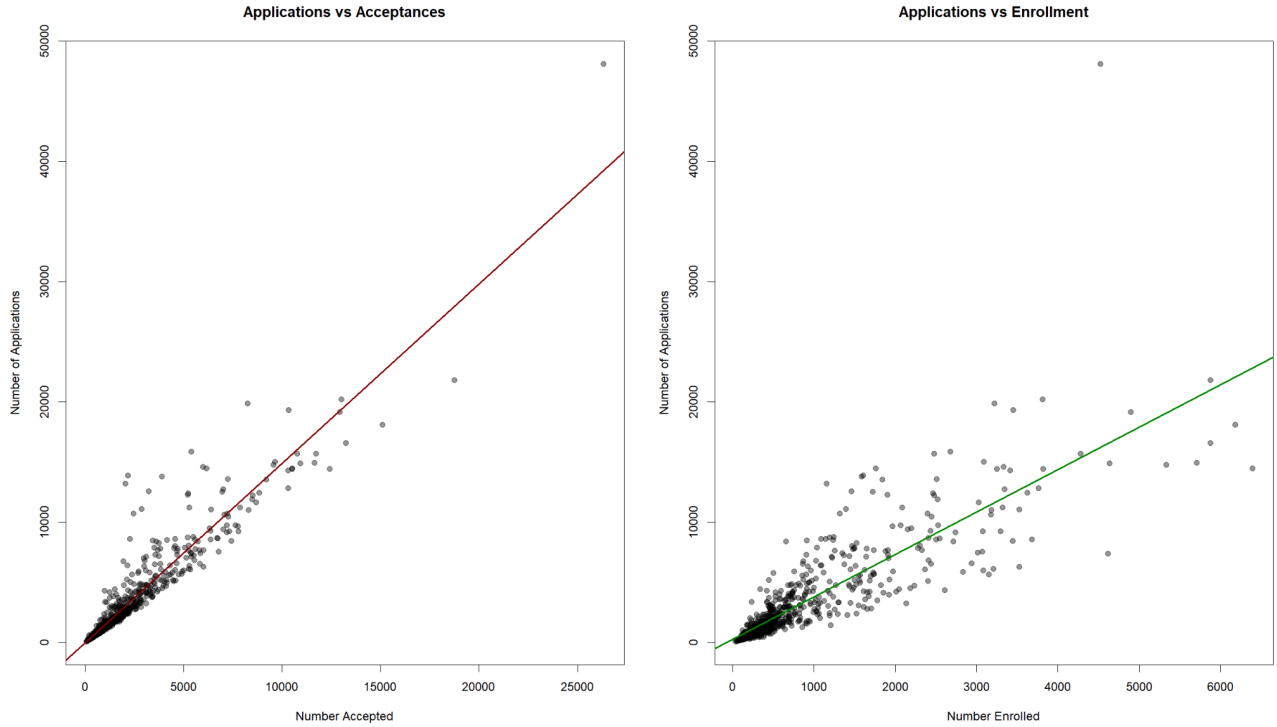

Figure 3. *Applications vs Acceptance and Enrollments.*

There is a positive correlation between the number of applications sent and number of applications accepted. It is easier to see that there is an outlier college that accepted over 25000 applicants and received the highest number of applicants. Most colleges will be accepting up to 5000 applicants for every 7500 applications. For enrollments, the outlier college with the most amount of applications does not have the most number of enrollments. There is a positive correlation with a less steep slope, as around 1000 students are enrolled for the 5000 applications sent.

From the EDA, it is shown that the number of applications a college receives is strongly correlated with how many students are accepted and then enrolled, as well as the total number of full-time undergraduate students. While these are very surface level observations, by removing them from model analysis more varied results can be obtained. Qualifications of faculty is another notable pair of variables, as better curriculum and higher education can lead to more students applying for those colleges. These relationships will be explored further as models are created and analyzed.

Methodology/Fitting Models

Linear regression

To analyze the relationship between Apps and the other variables in the College dataset, a least squares regression model is used. Linear regression models the expected value of a response variable as a linear function of predictor values (James et al., 2023). The general form of the multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

This is the first model used for this type of predictive analysis, as it assumes a linear relationship between the predictors and response. Using least squares, the relationship between application count and the other predictors are estimated by minimizing the sum of squared residuals, also known as the differences between observed and predicted values. The coefficients are estimated by:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

From the linear model, the LOOCV estimate of test error is to be found. Leave One Out Cross Validation evaluates by removing one observation, fitting the model on the remaining observations, then predicting the removed observation and repeating. The LOOCV estimate of test MSE is computed as:

$$\mathrm{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i.$$

The mean squared error which is being reported back is the expected prediction error. At first, a linear model was created using all variables as predictors to Apps. Next, LOOCV cross validation was performed on the model in order to calculate the estimated error of the model. After considering the EDA, specifically the pattern of Accept, Enroll, and F.Undergrad being prominent variables, a second model was created. This model does not include those three variables, allowing more focus on the other predictors while keeping the original analysis intact. This model will be used in the following sections in order to compare further correlation between predictors that can be hidden without removing the aforementioned variables.

Regression tree

While a linear regression model gives a good baseline, it assumes the relationship between variables is linear. Other predictors can affect the number of applications in more complex ways, and interactions between variables may exist that cannot be viewed through just a scatter plot or even a correlation matrix. In order to address this issue, a regression tree model is used. Unlike linear regression a tree does not impose a linear relationship assumption across the observations. Instead, it splits the predictors into regions based on minimum overall error (James et al., 2023). Within each region, the model predicts the average number of applications based on the effects of those specific predictors:

$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2,$$

First, the tree is constructed by recursively selecting predictors and split points that minimize the residual sum of squares. Each internal node represents a decision rule based on a predictor variable. These nodes are defined by rules, for example tuition being above or below a certain threshold, and then separated into a different branch. The prediction for each region is the mean number of applications for colleges that satisfy those rules. This is useful for analyzing nonlinear relationships, specifically in determining interactions between predictors. This process was applied twice similar to the previous section, once for the standard model and once for reduced model. A tree was created using the college dataset, then plotted with text for legibility.

Tree pruning

While the regression tree is a valid approach to model nonlinear interactions and relationships, a fully grown tree can become overly complex. The tree can emphasize random noise in the data, which takes away from the meaningful patterns that can be found. This is known as overfitting, and while an overfit tree can have a low training error, it can perform poorly in predicting new observations (James et al., 2023). In order to counteract overfitting, cost complexity pruning is performed. What pruning does is reduce the size of the tree by removing splits that add little to the overall accuracy. Instead of the algorithm growing into the largest tree, a sequence of smaller subtrees is created. Pruning introduces a complexity penalty parameter $\alpha \geq 0$ and selects the subtree that minimizes the cost complexity criterion:

$$\sum_{m=1}^{|T|} \sum_{i:\ x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

Each subtree is defined by the number of terminal nodes, and represents a different level of complexity. For each subtree, the prediction error is computed and the tree with the lowest cross validated mean squared error is selected as the optimal pruned tree. This pruned tree is compared to the original tree by looking at the mean squared error. If pruning reduces the cross validated error, then the original tree is overfitted. Even if the predictive performance is similar, the smaller tree is preferred due to it being easier to interpret as well as being more stable. If both the pruned and unpruned trees are similar or the same, it means that pruning does not improve performance.

Since pruning removes splits, if the optimal subtree was selected by cross validation then the tree's complexity was fine for the data. For the models used for College analysis, cross validation is used on both models. The best tree size is found by identifying the number of terminal nodes that correspond to the minimum cross validated deviance. In this case, the smallest estimated cross validated error occurred at 14 nodes, so the tree is pruned to this size. After the pruning, predictions are generated for the dataset and the MSE was calculated.

Bagging

The fifth methodology used in the analysis of the College data was Bagging. Standard decision trees succumb to high variance as the data set is being split in half at random and highly different results may be obtained from each half of the data. Linear regression in general tends to have lower variance so long as the ratio of n to p is moderately large. Bagging takes the strength of low variance found via linear regression and pairs it with the general methodology of decision tree construction. With Bagging, the data set is split into B repeated samples from the (single) training set, trains on the *bth* bootstrapped training set, and averages all the predictions (James et al., 2023). This is the general form of how Bagging operates. Applied to regression, B regression trees are created using B bootstrapped training sets and attaining the average of the prediction. As mentioned before, each of these individual trees has a high variance and low bias, but averaging these trees using regression greatly reduces the variance.

In the employment of Bagging, B is first set to 500, meaning that 500 separate regression trees are created and then averaged. 70% of the data was used as training data here and once the model was made, the variables for Y-hat and the testing data were created. The y-hat variable allows for making a prediction on the test data using the model, where this testing data is the 30% of the data not used to train the model. Taking the mean of $(\hat{y} - test)^2$, gives a way to see the MSE of the model. This step is then repeated with B set to 1000, or, 1000 separate regression trees averaged.

While we can easily plot the y-hat variable against the test data, plotting the model itself in a simple way is not possible. Recall that each individual regression tree constitutes its own decision tree plot. The model is more easily represented and interpretable as a Variable Importance plot. This plot allows us to see the % of importance each individual predictor within the data set contributes to the response variable, Apps.

Random forest

The Random Forest methodology is similar to bagging in many ways with one major distinction. Recall that when a regression tree is constructed the intrinsic property is the way in which the predictions are split into different nodes based on predictor values. With Bagging, each tree will select a number of predictors to include within the data set for the top split. When, as is the case with the College dataset, there is a single very strong predictor, most or all of the trees will include that predictor. This can cause many of the trees that are constructed via Bagging to look highly similar to one another as they are so strongly correlated.

Enter the major distinction. When the regression trees are constructed via the Random Forest methodology, a random sample of *m* predictors are chosen as candidates for the top split from the pool of *p* predictors. This random predictor selection fundamentally decorellates the trees as that single most important predictor is not necessarily included as in the case of Bagging. This means that, "...on average $(p-m)/p$ of the splits…" (James et al., 2023) will not even consider the heavily influential predictor in tree construction. The fundamental issue within Bagging is that despite the model being constructed of hundreds if not thousands of regression trees, with all the trees being highly correlated there will not be as much variance reduction as would be expected.

Utilizing the Random Forest approach overcomes this intrinsic weakness present within Bagging by ensuring there is little correlation tree to tree. That key variable, *m*, is set to 6 in the construction of the model as $p/3$ is the standard for use in regression and the *p* within the College dataset is 18. This means that at each top split within the regression tree construction, 6 random variables are selected from the total 18 available. As was the case with Bagging, 500 regression trees are constructed and then averaged, with the same y-hat and test variables created and in the same manner as before. The same as bagging as well, there is no easy way to plot a model that consists of 500 separate regression trees, so the Variable Importance plot is used to relay the individual importance of the predictors. The number of trees is then increased to 1000, same as with Bagging.

There are a couple of important trade-offs in deploying the Random Forest algorithm versus Bagging. Due to the nature of the way Random Forest selects random predictors, it can potentially lead to higher bias within the model. In addition, Random Forest is often more computationally expensive than bagging and can potentially require more memory.

Results

Linear regression

Two linear regression models were developed to predict the number of college applications using ordinary least squares estimation, as implemented by the glm() function in R.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -445.08413  408.32855  -1.090 0.276053
PrivateYes  -494.14897  137.81191  -3.586 0.000358 ***
Accept         1.58581    0.04074  38.924  < 2e-16 ***
Enroll        -0.88069    0.18596  -4.736 2.60e-06 ***
Top10perc     49.92628    5.57824   8.950  < 2e-16 ***
Top25perc    -14.23448    4.47914  -3.178 0.001543 **
F.Undergrad    0.05739    0.03271   1.754 0.079785 .
P.Undergrad    0.04445    0.03214   1.383 0.167114
Outstate      -0.08587    0.01906  -4.506 7.64e-06 ***
Room.Board     0.15103    0.04829   3.127 0.001832 **
Books          0.02090    0.23841   0.088 0.930175
Personal       0.03110    0.06308   0.493 0.622060
PhD           -8.67850    4.63814  -1.871 0.061714 .
Terminal      -3.33066    5.09494  -0.654 0.513492
S.F.Ratio     15.38961   13.00622   1.183 0.237081
perc.alumni    0.17867    4.10230   0.044 0.965273
Expend         0.07790    0.01235   6.308 4.79e-10 ***
Grad.Rate      8.66763    2.94893   2.939 0.003390 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -4.582e+03  1.074e+03  -4.267 2.23e-05 ***
PrivateYes  -3.508e+03  3.396e+02 -10.329  < 2e-16 ***
Top10perc    3.615e+01  1.470e+01   2.459  0.01415 *
Top25perc    1.902e+01  1.184e+01   1.606  0.10860
P.Undergrad  5.956e-01  7.780e-02   7.655 5.86e-14 ***
Outstate     3.487e-02  5.010e-02   0.696  0.48668
Room.Board   3.612e-01  1.277e-01   2.828  0.00480 **
Books        6.493e-01  6.357e-01   1.021  0.30741
Personal     2.566e-01  1.666e-01   1.540  0.12393
PhD          9.762e-01  1.237e+01   0.079  0.93711
Terminal     2.745e+00  1.357e+01   0.202  0.83970
S.F.Ratio    1.077e+02  3.444e+01   3.127  0.00183 **
perc.alumni -4.290e+01  1.076e+01  -3.986 7.38e-05 ***
Expend       1.424e-01  3.285e-02   4.335 1.65e-05 ***
Grad.Rate    4.157e+01  7.758e+00   5.359 1.11e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 4. *Summary Statistics for the full linear regression model and reduced linear regression model. The following predictors were statistically significant at the 0.05 level: Accept, Enroll, Top10perc, Top25perc, PrivateYes, Outstate, Room.Board, Expend, and Grad.Rate.*

Of these, Accept was the most important, having a t-statistic of 38.924 and a regression coefficient of 1.586. This indicates that for each additional student accepted, the predicted number of college applications increases by 1.586, holding other predictors constant. The negative regression coefficient for the Enroll variable, -0.881, indicates a negative relationship between the predictors and the response variable, as might be expected given the strong correlation between the predictors. The predictors of Books, Personal, Terminal, S.F.Ratio, and perc.alumni were not significant and did little to improve the model.

The residual deviance for the full model was 823,060,000 on 759 degrees of freedom, so the dispersion parameter was 1,084,400. The ratio of the residual deviance to the null deviance for the full model is approximately 0.071, meaning that the model explains approximately 92.9% of the total deviance. As the right skewness of the Apps variable was identified in the EDA, a level of heteroskedasticity is expected among the residuals, particularly at higher levels of prediction where the impact of the small number of large institutions is felt.

Using the cv.glm() function, the LOOCV estimate of the test MSE of the full model is:

$$CV_{(n)} = 1{,}276{,}987$$

A second model was developed after removing the Accept, Enroll, and F.Undergrad variables. As a result of the removal of the aforementioned variables, the significant predictors changed considerably. In this case, the significant predictors were identified as PrivateYes, P.Undergrad, Room.Board, S.F.Ratio, perc.alumni, Expend, and Grad.Rate. These variables demonstrated significant relationships that were initially masked by the dominant size-based predictors. As a result of the removal of the significant predictors, the LOOCV test MSE value of the reduced model is significantly higher at:

$$CV_{(n)} = 7{,}997{,}698$$

This is approximately 6.25 times the full model's error value, which reveals that the majority of the predictive information is contained within the Accept, Enroll, and F.Undergrad variables.

Regression tree

For the regression trees two trees were created: one based on the original model, and another using the model without Accept, Enroll, and F.Undergrad. The tree algorithm for the first model determined that only the number of students accepted and the percentage of incoming students from the top ten percent of their high school class were needed to minimize the prediction error. This means that these two variables contribute most of the variation in application counts, and that the other variables did not improve the model enough to create splits.

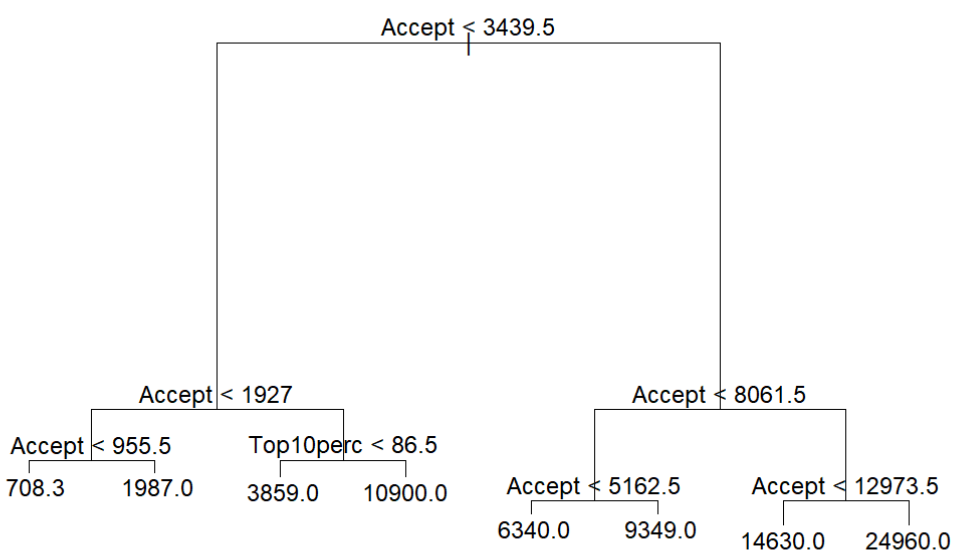

Figure 5. *Regression tree with original model. Accept takes up the structure of the tree, supporting the EDA's findings of having a strong relationship with Apps.*

There are 8 terminal nodes, each representing a set of colleges satisfying conditions based on Accept and Top10perc. The initial split happens at Accept > 3439.5, separating the smaller colleges from the larger ones. Further splits separate colleges based on Accept thresholds as seen with < 1927, 8061.5, etc. As Accept increases, predicted application counts rise as well. The only additional predictor used was Top10perc, which appeared in one split, meaning that academic selectivity contributes predictive value in a specific range of college sizes. The residual mean deviance was 2,031,000 which indicates a strong sample fit.

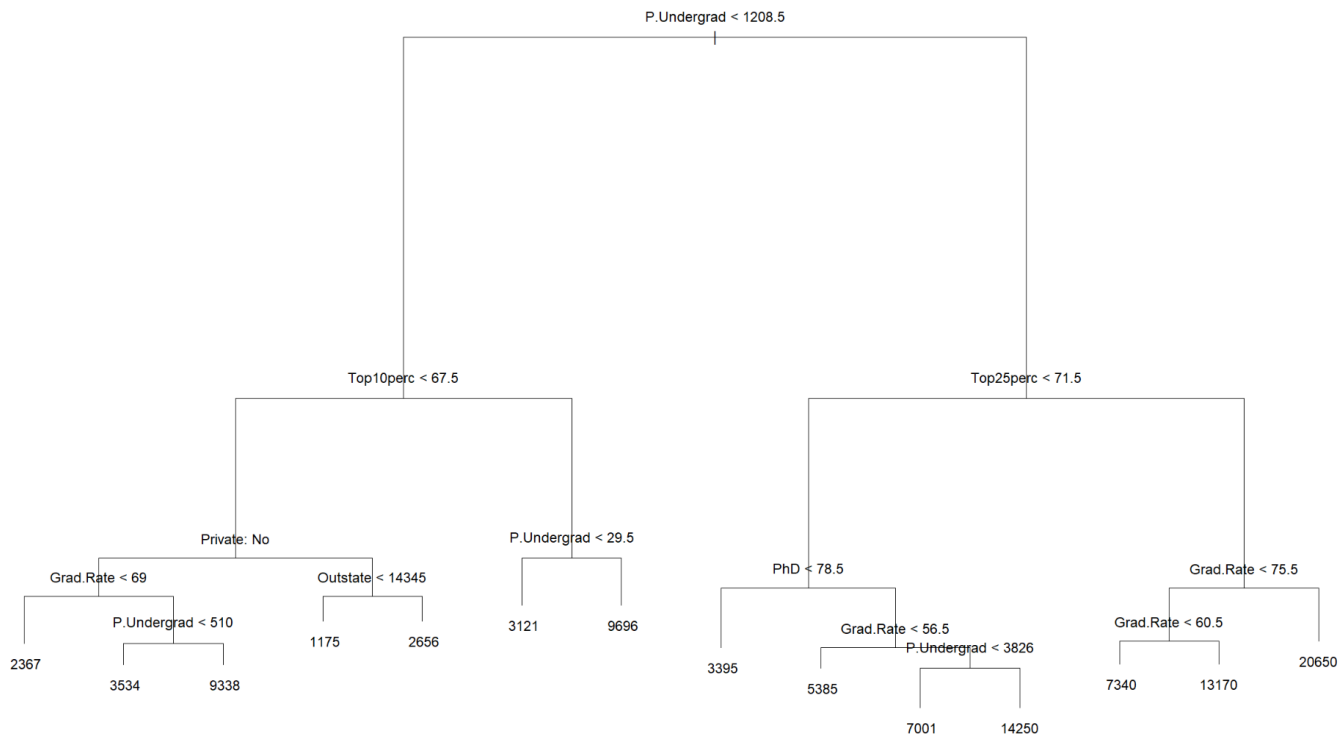

Figure 6. *Regression tree with reduced model. A second tree is fitted after removing Accept, Enroll, and F.Undergrad in order to look at other relationships present in the data.*

The resulting tree ended up being more complex, with 14 nodes and seven predictors: P.Undergrad, Top10perc, Top25perc, Grad.Rate, Outstate, Private, and PHD. The first split occurs at P.Undergrad, which can lead to the assumption that part time undergraduates enrolling is a major predictor when Accept is removed. The further splits share the pattern of being measures of academic content and qualifications, which can contribute to more students applying to those specific colleges. In the removal of population size based predictors, the model relies on the institution's quality of education to explain application volume. However, the residual mean deviance increases to 5,555,000, meaning that this particular model does have a poorer fit compared to the full model.

After comparing these two trees, we can conclude that Accept is the most influential predictor of application count. When included, the tree is simpler and has a better fit with fewer splits. When removed, the model is more complex and performs worse. This highlights the importance of size in explaining variation in applications.

Tree pruning

Cost complexity pruning is used to see if reducing tree complexity improves prediction performance. This is applied by using cv.tree() to evaluate subtrees of different sizes, where each subtree is associated with a cost complexity parameter α\alpha α that penalizes tree size. The optimal subtree is selected by minimizing the cross validated deviance across all candidate values of α. The cross validation results are summarized below:

Terminal Nodes	α	CV Deviance
8	$-\infty$	2,589,585,642
7	202,873,515	3,071,398,626
6	228,050,501	3,077,779,403
5	282,105,525	3,284,312,194
4	426,616,632	3,334,919,547
3	893,815,939	3,953,590,418
2	1,596,851,915	5,769,068,298
1	6,430,760,151	11,650,525,253

Table 4. *Cross Validation Results*

The minimum cross validated deviance of 2,589,585,642 occurs at 8 terminal nodes with $\alpha = -\infty$, indicating that no pruning penalty is required to select the optimal tree and the full tree is already optimal. The training mean squared error for both the pruned and unpruned trees was 2,010,567. This means that the full tree already represents the optimal level of complexity according to cross validation.Since the cross validated optimal size equals the original size and the MSE values are identical, there is no evidence of overfitting.

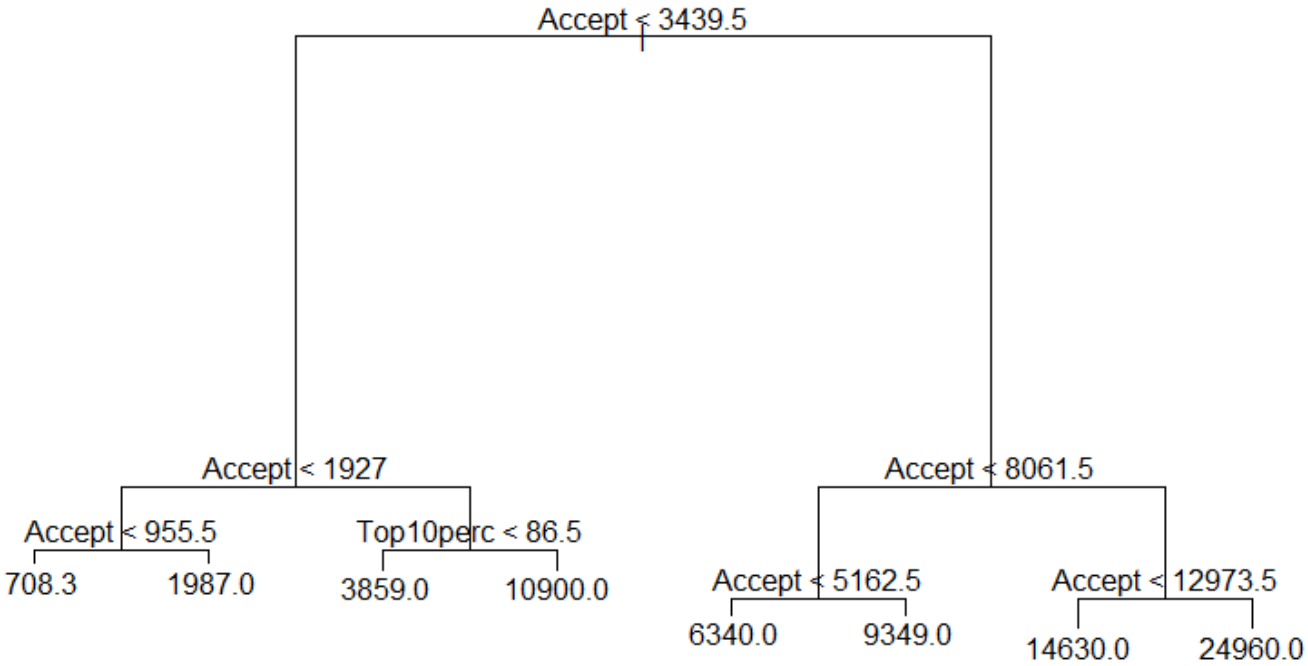

Figure 7. *Pruned Tree of normal model.*

This tree is the exact same as the unpruned tree from the previous section. This indicates that the original tree doesn't require pruning to reach full optimization for the original model. The optimal size chosen by minimizing the cross validated deviance is 8 terminal nodes, which led to the tree being pruned to this size. Since the MSE values for the pruned and unpruned trees are the same, this means that there is no overfitting in the original tree, which implies that the original tree complexity is already appropriate.

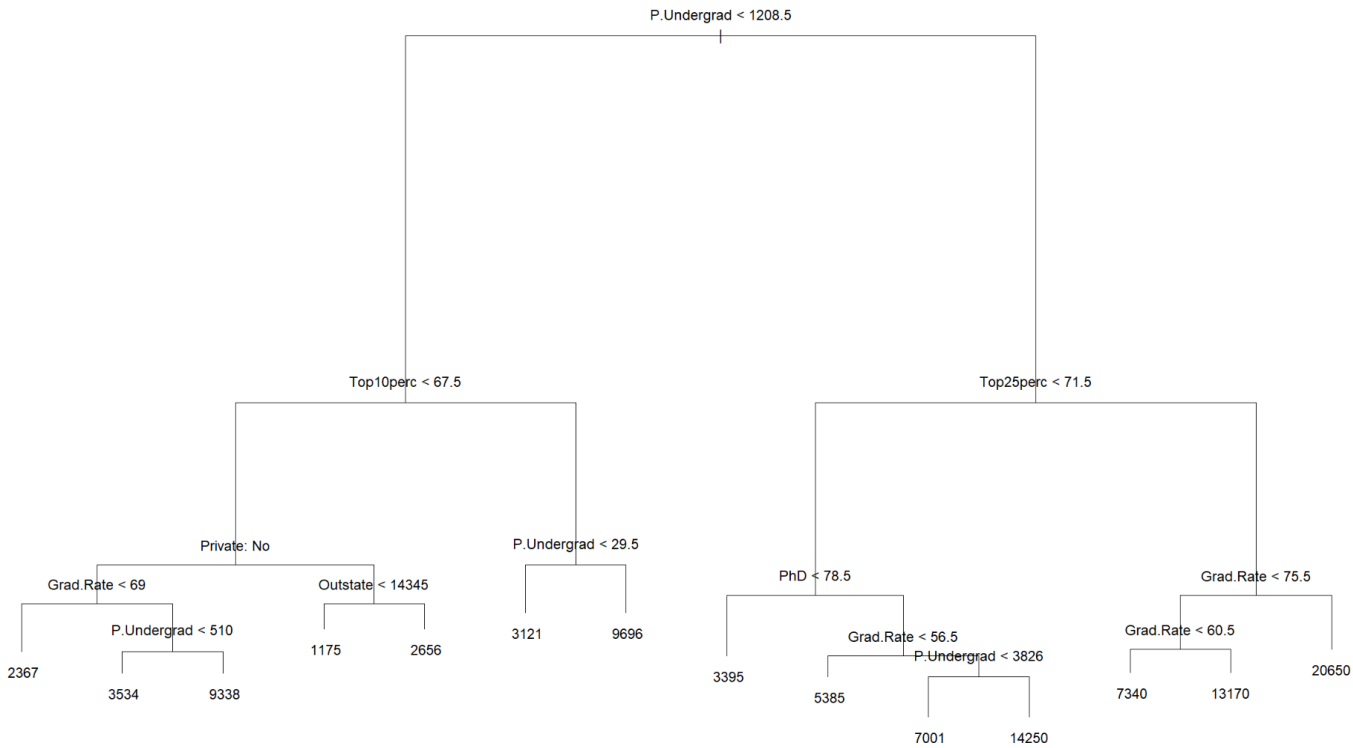

Figure 8. *Pruned Tree of reduced model. For the reduced model, the optimal size that gave the minimum prediction error was 14 terminal nodes, after which the tree was pruned to the optimal size.*

The MSE values for the pruned and unpruned were the same at 5,454,741, meaning that the regression tree did not improve after pruning. These values show that the MSE values for the full model are much lower than those obtained using the reduced model, which implies that the full model has a better fit than the reduced model.

This means that the variables were removed: Accept, Enroll, and F.Undergrad have a lot to do with the prediction process. Based on the above analysis, the full model is better than the reduced model because its MSE values are much lower than those obtained using the reduced model. The variables that appeared at the top of the full tree, such as Accept, are the most important predictors because they produce the greatest decrease in residual deviance.

Bagging

To examine the Bagging models, the first plots created provide deeper insight into the predictive accuracy of the model via plotting the predicted number of applications versus the actual number of applications.

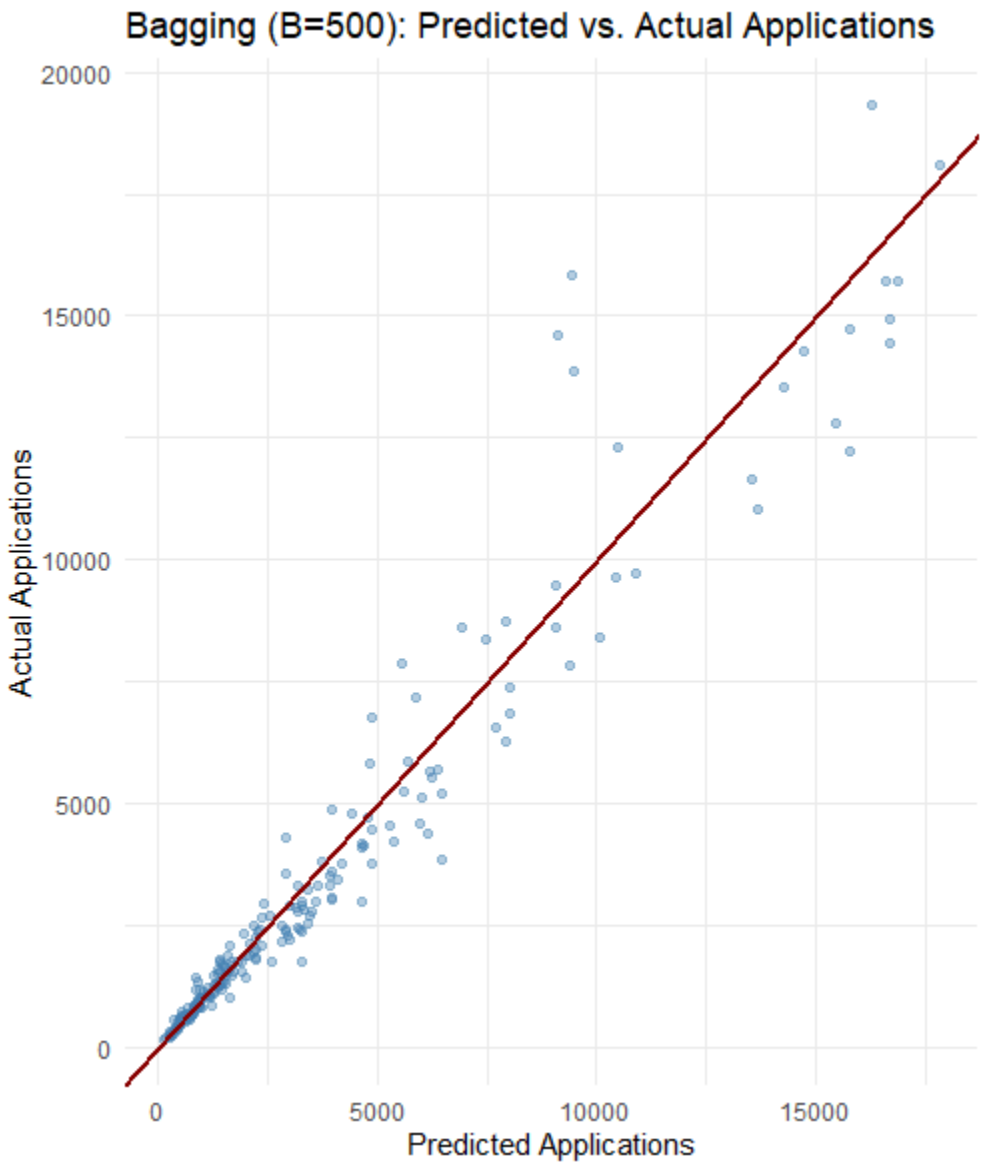

Figure 9. *Predicted vs. actual number of applications for Bagging model with B = 500. The red line represents the perfect prediction with slope = 1 and intercept = 0.*

As you can see in figure 9, this model performs well at lower numbers of applications and begins to lose some accuracy as the number of applications increases. This is in line with Figure 1 in the EDA where it is noted that a few universities receive and accept a massive number of applications, and as such have increased variance creating heteroskedasticity in the model.

As the number of applications get up to around 12,000, the model consistently underpredicts the number of applications. This shows Bagging as a methodology is struggling with the more extreme values in the data set and since they are more rare in the training data, the average tree regresses to the more commonly found values, i.e, lower number of applications.

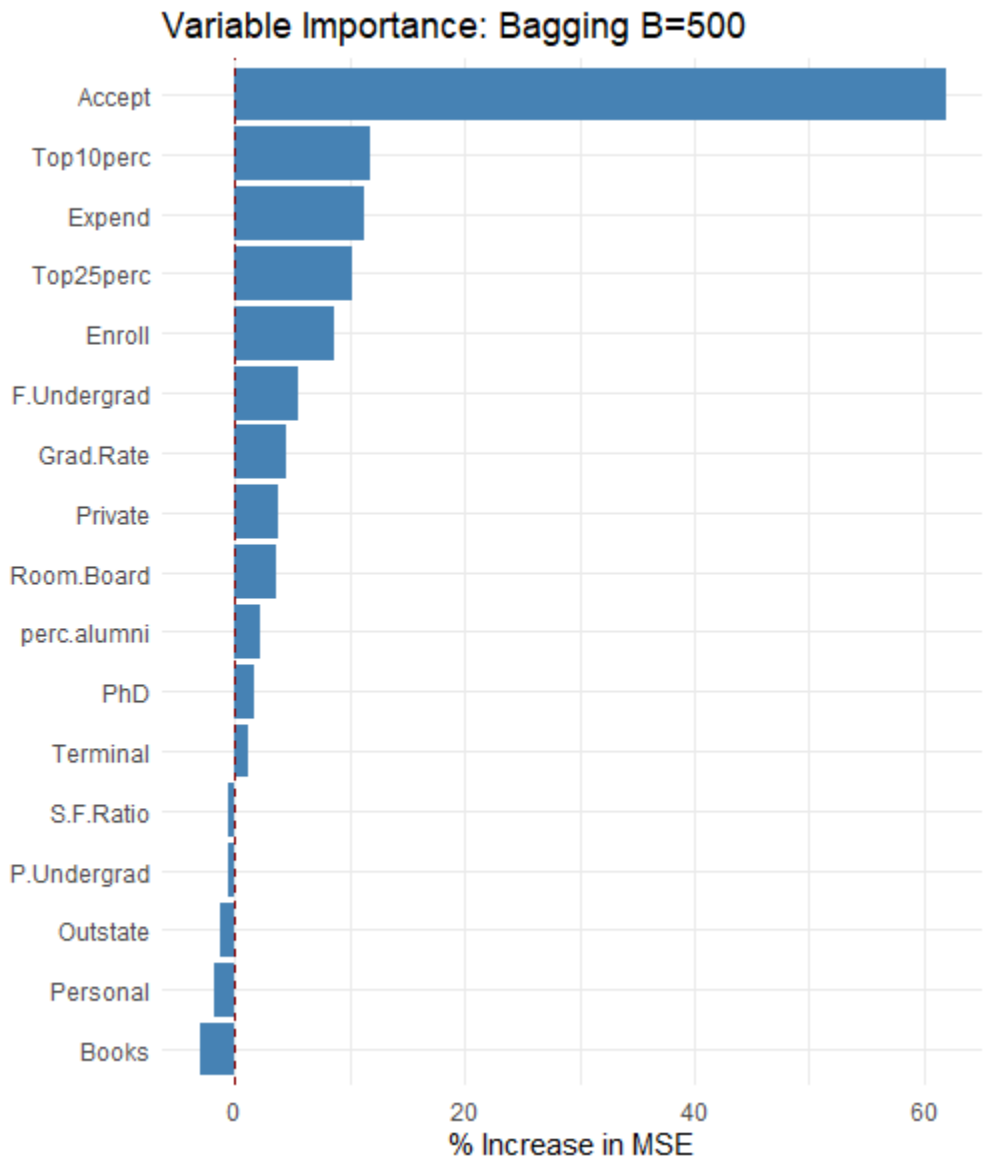

Figure 10. *Percentage increase in Mean Squared Error (MSE) within Bagging model where B = 500 for each individual predictor. Larger values play a greater level of importance in the model's prediction accuracy.*

As evidenced here in figure 10, the Accept predictor has a hugely influential role within the model compared to all other predictors with a % MSE increase of over 60%. This means that if Accept is not included as a parameter, the MSE of this model would increase by 60%, making it far less accurate. This makes intuitive sense as the number of students accepted into a university is directly linked to the

number of applications the university receives.

Beneath Accept in level of importance within the model; Top10perc, Expend, Top25perc, and Enroll all have an % MSE increase of around 10%. This shows that they individually have played some meaningful part in the model's overall accuracy but are far less critical for prediction accuracy than the Accept predictor.

Further down the chain of predictors we can see some contribution to % MSE increase for Grad.Rate, Private, Room.Board, perc.alumni, PhD, and Terminal, but none of these predictors are significant. The predictors Books, Personal, and Outstate do not help the model's predictive accuracy whatsoever.

The standard when implementing Bagging as a methodology at first is to set the number of regression trees created to 500 like we have seen in this model. It is worth investigating whether or not increasing the number of regression trees from 500 to 1000 will resolve any normality issues within the model.

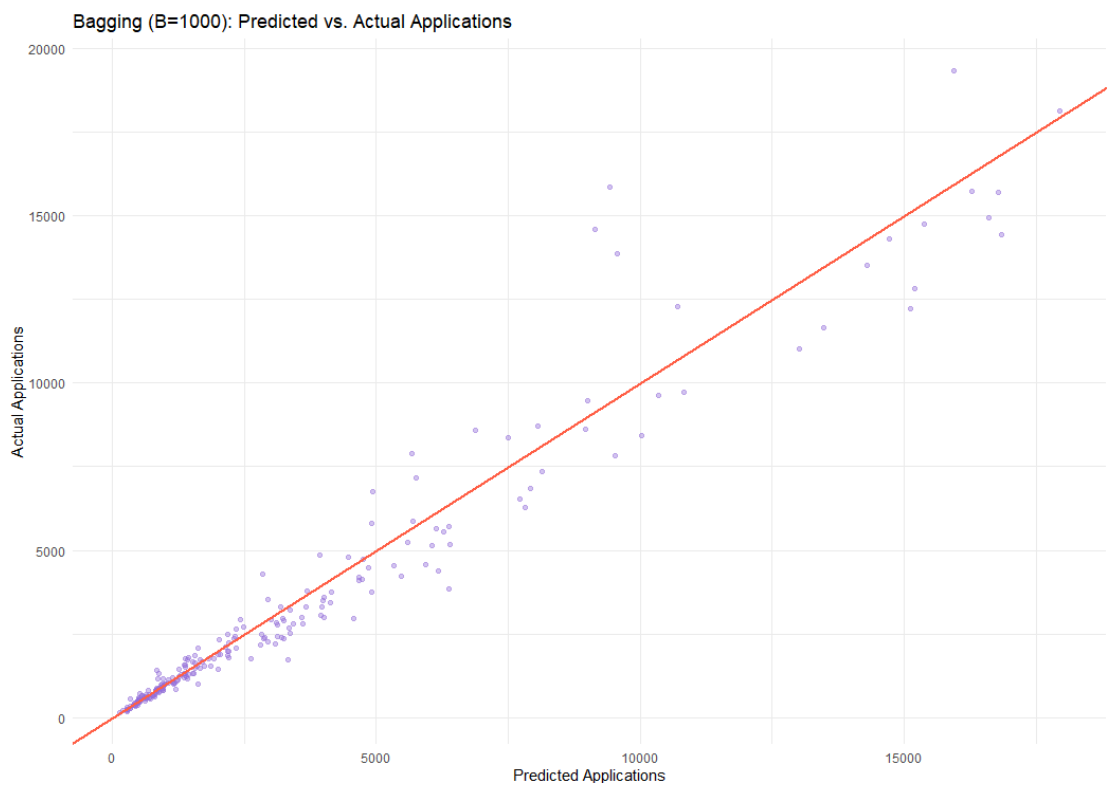

Figure 11. *Predicted vs. actual number of applications for Bagging model with B = 1000. The orange line represents the perfect prediction with slope = 1 and intercept = 0.*

Figure 11 looks markedly similar to figure 9 in that its predictive accuracy is at first extremely strong with a large cluster about the perfect prediction line at lower numbers of applications, and then begins to lose accuracy as the application count increases. Once again, like when B = 500, there is some heteroskedasticity present within the model at a higher number of applications as the points fan out and the model repeatedly underpredicts the application number. This suggests that by 500 regression trees

implemented the normality assumptions of the model are quite strongly in place. It is worth noting however that in figure 11 when B = 1000 those higher value predictions, while underpredicting, are marginally closer to the diagonal than in figure 9 when B = 500. In addition, within the mid range of the model, the predictions are clustered slightly close to the diagonal as well. Further investigation is needed to see if there are any outstanding differences with the increase in regression tree creation to the normality assumptions.

Variable importance, much as was the case with B = 500 Bagging and indeed other models in the analysis, plays a huge role in understanding just how exactly the predictors are playing a part within the model's predictive accuracy capabilities.

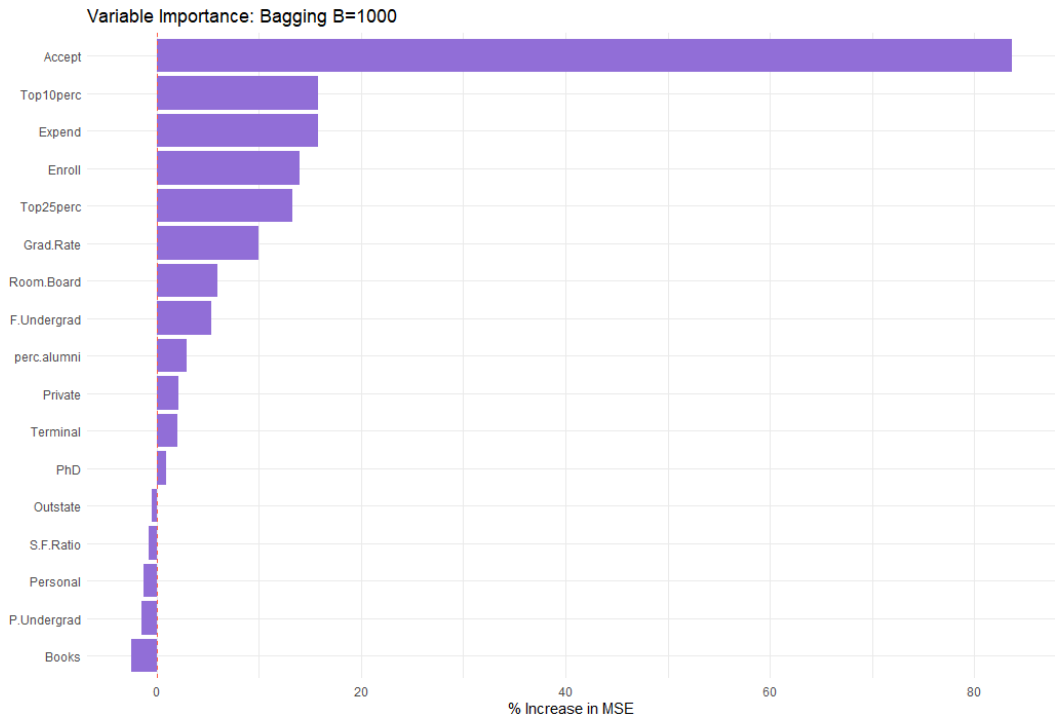

Figure 12. *Percentage increase in Mean Squared Error (MSE) within Bagging model where B = 1000 for each individual predictor. Larger values play a greater level of importance in the model's prediction accuracy.*

There are some notable differences in the variable importance plots from figure 10 where B = 500 and here in figure 12 where B = 1000. The most significant difference is that the importance of the Accept predictor has become even more pronounced with the inclusion of an additional 500 regression trees. This confirms that as the Bagging methodology increases in complexity, the model becomes even more certain that the Accept predictor is the overwhelmingly dominant predictor within the dataset.

The ordering of the other predictors has also changed as Expend has moved up in % MSE increase to sit just about equal with Top10perc while Enroll has dropped slightly. Grad.Rate and Room.Board also see a marked increase in % MSE with more trees added, declaring themselves as more important predictors when model complexity increases.

Random forest

With the analysis of the Bagging methodology, figures 9-12 gave great insight into the models, the normality assumptions therein, and how the predictors were meaningfully contributing to accuracy. These plots also work exceptionally well for Random Forest and the fundamental way tree construction works is so similar between the two methodologies. Recall that the major distinction between these methodologies is the *mtry* variable, which is responsible for selecting the number of predictors to include within each regression tree constructed. In Bagging all available predictors are used, where in random forest *p/3* predictors are used, where *p* in this case = 18.

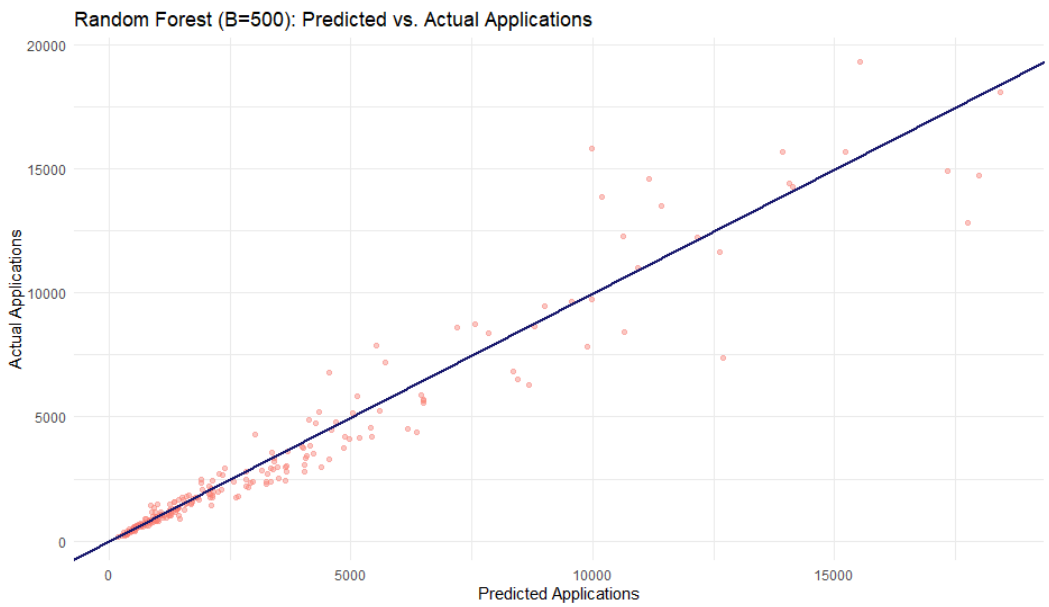

Figure 13. *Predicted vs. actual number of applications for Random Forest model with ntree = 500. The purple line represents the perfect prediction with slope = 1 and intercept = 0.*

There are some clear and noticeable differences in figure 16 when compared with the figures 9 and 11 from the Bagging results. Understanding that the consistent under prediction around the larger values in Bagging came from the influence of the Accept predictor, the nature of Random Forest construction means that some amount of trees do not include that highly important predictor. This could account for the seemingly high dispersion both above and below the prediction line at higher values along the diagonal. When measuring the MSE of the Bagging models and Random Forest models, the Random Forest models have an MSE nearly double that of the Bagging models. This is showcased here in the breadth of dispersion present. Overall, there is a greater spread in the predictions and seemingly slightly less heteroscedasticity present here, yet it still remains.

It was clear with the variable importance plots created for Bagging that the Accept predictor was incredibly important to the model's prediction accuracy and accounted for a massive amount of the %MSE increase.

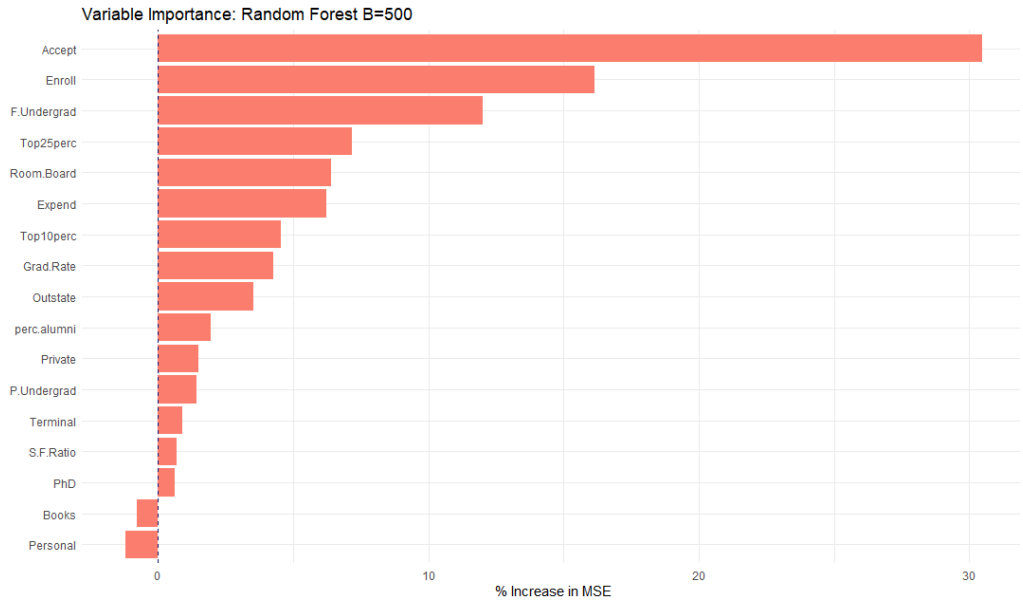

Figure 14. *Percentage increase in Mean Squared Error (MSE) within Random Forest model where ntree = 500 for each individual predictor. Larger values play a greater level of importance in the model's prediction accuracy.*

There is right away a noticeable difference in the variable importance within the Random Forest methodology compared to Bagging. Accept still dominates as the most important predictor with the dataset but has a % MSE increase of around half of that of the Bagging model. This makes sense from what we understand about the fundamental difference in model construction between the two methodologies as Accept is sometimes not present with the regression tree construction in Random Forest. Predictors are much more spread out within Random Forest in % MSE increase here as well. Enroll plays a hugely important role in model prediction followed closely by F.Undergrad. Random Forest is clearly using more of the predictors than the bagging model, and it's not only a reduction in the % MSE increase for Accept that's noteworthy but indeed a dispersion of that % MSE increase across the other predictors in the dataset. This is occurring to such a degree that the predictor Outstate even appears with some non-zero level of importance within the model for the first time.

Figure 15 provides us with a fantastic picture as to what happens when Accept isn't defacto included within nearly every regression tree, but seeing Bagging first helps to contextualize this as not a wholly accurate picture being painted as to which predictors play the most important part within the dataset.

So, what happens to these levels of importance when the number of trees implemented is doubled from 500 to 1000? Is the Accept variables % MSE increase going to become more pronounced as was seen in the Bagging models?

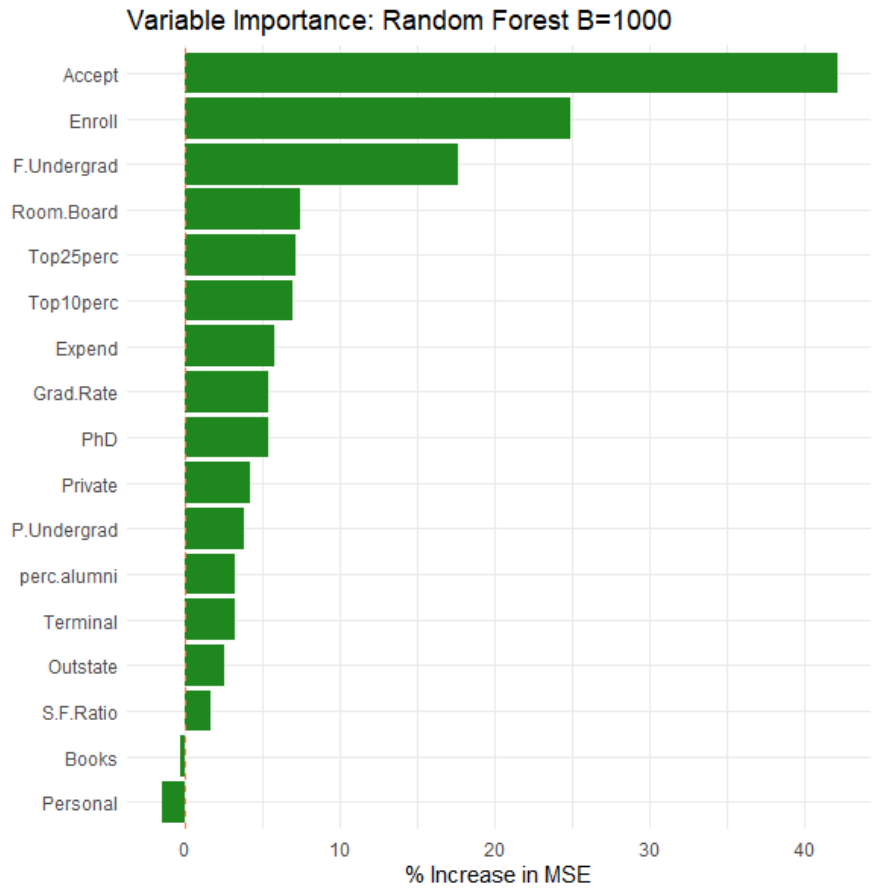

Figure 15. *Percentage increase in Mean Squared Error (MSE) within Random Forest model where ntree = 1000 for each individual predictor. Larger values play a greater level of importance in the model's prediction accuracy.*

Much as was the case with Bagging, with an increase in the number of regression trees comes an increase in the importance of the Accept predictor, jumping from just above 30% when ntree = 500 to just above 40% when ntree = 1000. This showcases the power of Accept within the dataset because even with the restriction in place due to the nature of tree construction within Random Forest, it stands out as the clear and prevailing predictor for model accuracy.

With the additional 500 regression trees there is also a spread of % MSE increase seen within the other predictors all across the board. The gap widens slightly between Enroll and F.Undergrad with both predictors playing an important role in model accuracy. This dispersion of importance seen before continues into the less important variables but not to such a degree there are any noteworthy predictors that were not present with 500 regression trees.

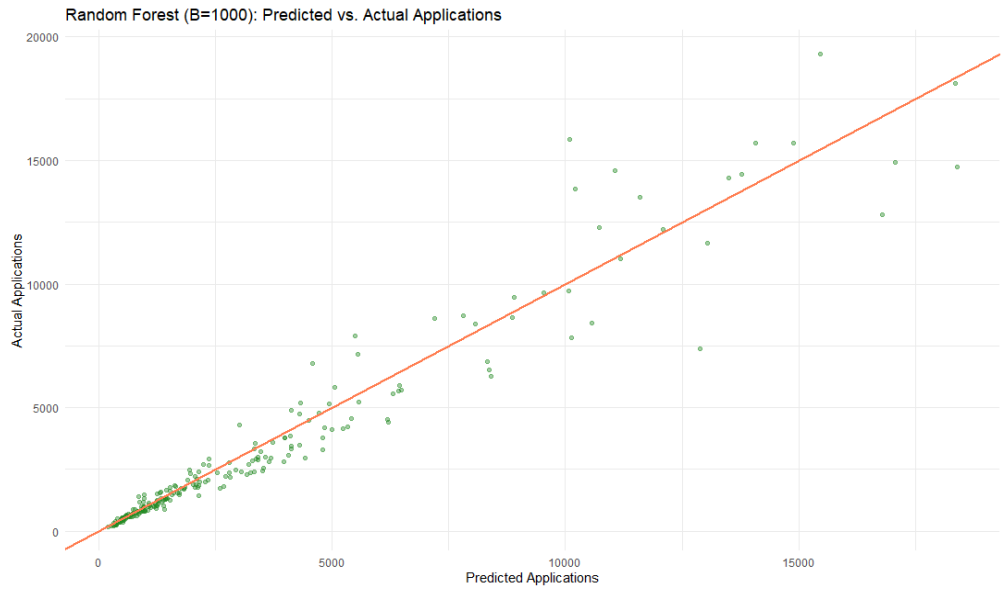

Figure 16. *Predicted vs. actual number of applications for Random Forest model with ntree = 1000. The orange line represents the perfect prediction with slope = 1 and intercept = 0.*

With an increase in regression trees from 500 to 1000 there is a more linear relationship that has evolved in the model, however there is still heteroskedasticity present. This seems to confirm that heteroskedasticity is simply a prevailing characteristic within the dataset due to the nature of a few universities having such extreme values in Apps.

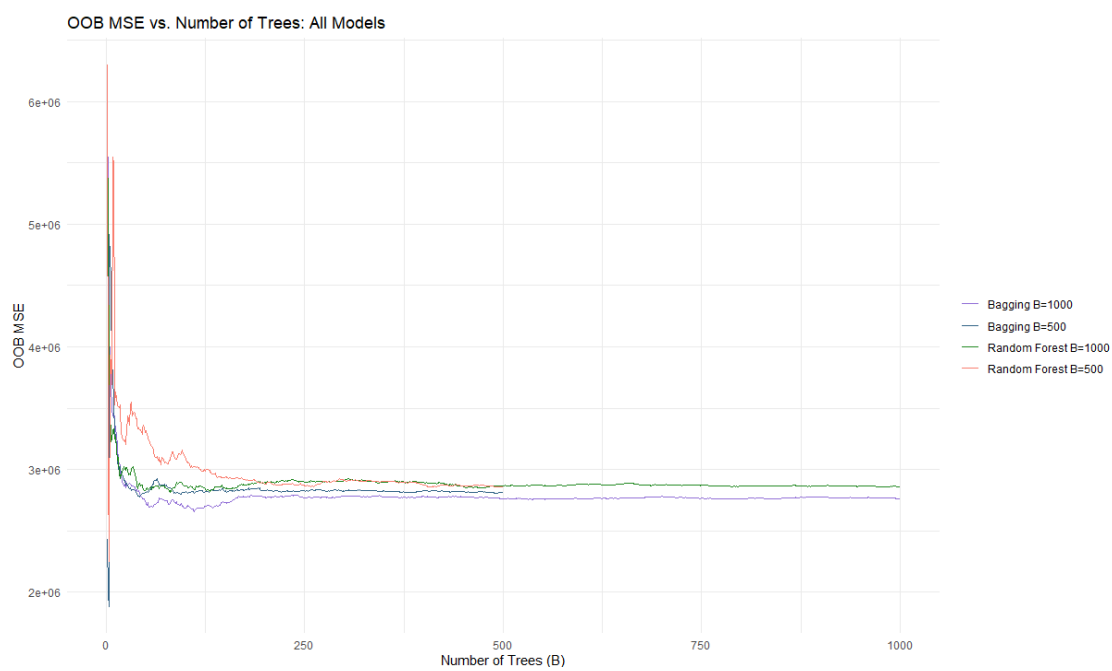

Figure 17. *The unused OOB "Out Of Bag" MSE for all Bagging and Random Forest models.*

Figure 24 provides excellent insight into the predictive accuracy of all of the Bagging and Random Forest models side by side. In both the Bagging and Random Forest methodologies, a number of bootstrap samples are taken with replacement, meaning that at each iteration of a new regression tree construction, some amount of the training data is not used. The OOB, or Out Of Bag error, takes the unused training data and calculates the MSE, giving a very strong way to validate the models.

The two Random Forest models show moderate instability at a lower level of regression trees which once again ties back to the intrinsic nature of tree construction within the methodology. When random variables are selected and the single most important predictor, Apps, is sometimes excluded, it leads to more volatility at these lower values. In all cases it seems as though the MSE stabilizes by the 250 tree mark. While there were interesting changes that occurred with the increase in regression trees created that led to analysis to deepen understanding of how these models function on a fundamental level, the increased complexity may not ultimately be necessary for the correct implementation of one of these models on the College dataset.

Comparison

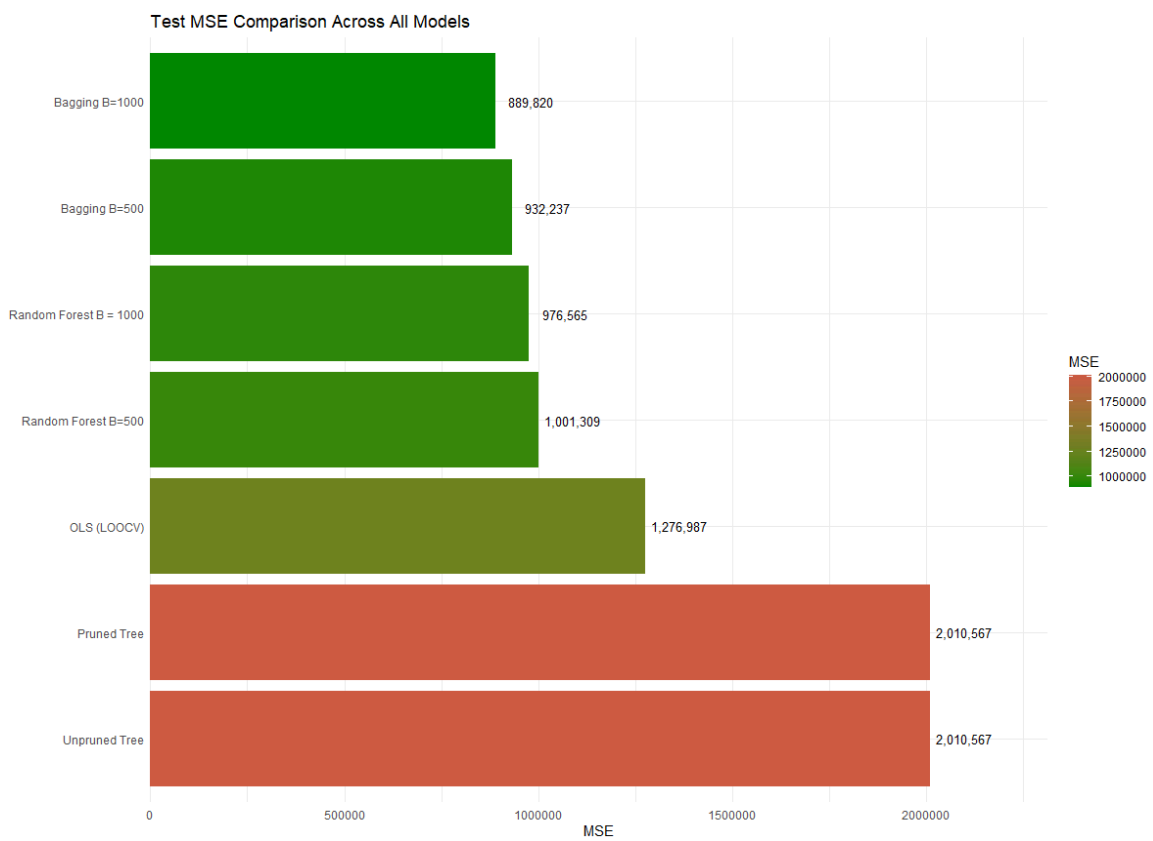

Figure 18. *Test MSE Comparison Across All Models. Bagging with B = 1000 has the lowest MSE, while the Unpruned Regression Tree has the highest.*

Random Forest is an extremely strong methodology for prediction accuracy generally speaking, but the results from the analysis show that while being computationally more expensive, this is not a tradeoff made for higher predictive accuracy. Both Bagging models implemented have a lower MSE, meaning higher accuracy, while being less intensive to implement. In addition, figure 18 clearly demonstrates that no real benefit occurs past approximately 250 regression trees as by that point, the MSE has stabilized. For predictive accuracy and computational ease of implementation, Bagging with ntree = 500 is the clear model choice between the 4 Bagging and Random Forest models.

Conclusion

This analysis compares different predictive modeling techniques for the number of applications using the College dataset. The techniques include linear regression, regression trees, bagging, and random forests. In all techniques, the variables related to institutional size have the greatest influence on the number of applications. Variables such as the number of students accepted and enrolled have substantial information regarding the demand for applications. These variables improve the predictive performance of the models.

The regression tree models, both pruned and unpruned, have the same mean squared errors of 2,010,567. This suggests that the model was not improved by the pruning process. In other words, the unpruned regression tree was not overfit. Although the regression tree can handle non-linearities and interaction effects, the accuracy of the model is lower than the other techniques. This suggests that the relationship in the data is mostly linear. In addition, the relationship is dominated by the scale variables. The linear regression model using Leave One Out Cross Validation performs better than the regression tree models. The MSE for this model is 1,276,987. This suggests that the linear model is appropriate for the data. This could be due to the fact that the number of applications increases proportionally with the institutional size variables.

The prediction error was reduced by using ensemble learning methods. The MSE for the random forest model was found to be 1,001,309 when B = 500 and 976,565 when B = 1000. The decrease in prediction error is because the random forest model reduces the variance in the model. However, the random forest model performed slightly poorer than the bagging model, possibly because the random forest model randomly chooses the predictor variables to split at each node, possibly weakening the impact of the dominant size-related predictor variables.

Of all the models, the bagging model performed the best in terms of test MSE, with values of 932,237 when B = 500 and 889,820 when B = 1000. The prediction error is reduced when the number of bootstrap samples is increased from 500 to 1000, as iterating through more trees improves the model's prediction accuracy. However, the decrease in prediction error is fairly small, given the increased computational burden in aggregating twice the number of trees in the model. Although B = 1000 has the lowest test MSE, B = 500 is still competitive, given that it has a lower computational burden. In conclusion, the bagging models at B = 500 and B = 1000 performed the best in prediction accuracy.

Recommendations

If the main goal of this analysis is maximizing the prediction accuracy, bagging with B = 1000 trees is preferred over B = 500 since it achieves the lowest mean squared error. The improvement shows the effectiveness of variance reduction. If computational efficiency is more important in that context, bagging with B = 500 is a better choice. The increase in MSE is a small difference which makes it a good compromise between accuracy and efficiency. The regression trees and linear regression models are not recommended for this dataset, as their performance was consistently worse than the bagging approach.

Future Improvements

There are some improvements that can be made to enhance the analysis further. Transformations on the response variables can be explored, given that the distribution of the applications is right skewed. The logarithmic transformation can be explored further to enhance the model fit and the stability of the variance. A consistent validation approach can be employed, given that linear regression employed LOOCV, while the ensemble methods employed training and testing data. A consistent validation approach can be employed for all the models, such as cross-validation, to enable a comparison between the models.

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